CFD interpretation practical: transonic airfoil screening

Instructions and reference material

You have **30 minutes**. Work individually. No calculator, network access, AI tool, code, or notes are permitted. Show a short calculation where requested; concise evidence-based statements are sufficient elsewhere.

Scenario

A 250 kg high-subsonic cargo UAV, including its payload, is screened at one flight condition. The results below come from a two-dimensional inviscid finite-volume airfoil calculation. Three candidate configurations have fixed thickness but different camber and incidence.

Reference relations

Use $g = 10$ N/kg and the supplied load factor $n = 1.2$:

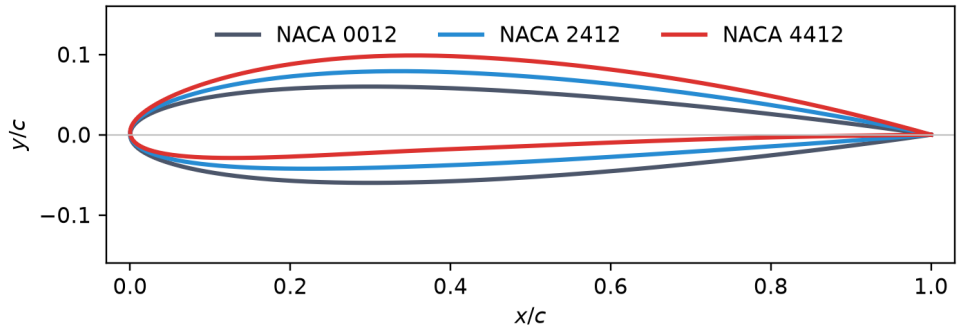
$$W = mg, \qquad L_{\text{req}} = nW, \qquad L = q_\infty S C_L, \qquad D = q_\infty S C_D.$$

For this problem, $q_\infty S = 10.0$ kN. The strong-shock screening limit is

$$M_{\text{surface,max}} \leq 1.30.$$

A rapid decrease in Mach number paired with a sharp pressure recovery in a surface C_p curve indicates a shock. A strong shock can raise transonic wave drag and may cause separation in a real viscous flow.
Model boundary. The CFD coefficients are used here as a simplified wing-screening approximation. They do not certify viscous drag, separation, three-dimensional effects, structural loads, stability, or full aircraft performance.

1. Requirements and preliminary recommendations [10 points]



Candidate	Camber	α	C_L	C_D	$M_{\text{surface,max}}$
A	high	2.5°	0.36	0.026	1.42
B	moderate	2.0°	0.32	0.015	1.24
C	low	2.0°	0.28	0.009	1.14

Engineer A: "Select A. It has the greatest lift coefficient and therefore the greatest payload margin."
Engineer B: "Select C. It has the smallest drag coefficient and therefore is the most efficient choice."

1. (2 points) Calculate L_{req} and $C_{L,\text{req}}$.

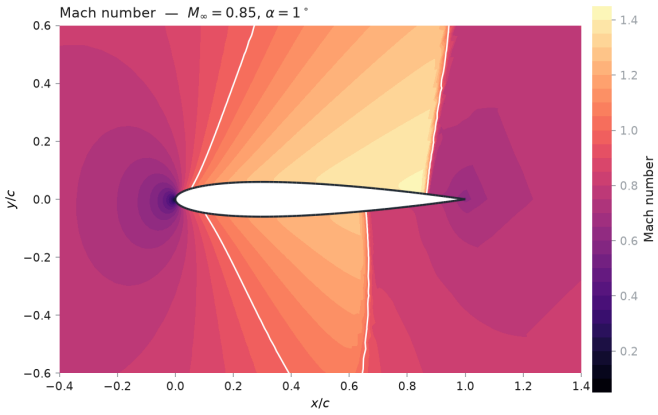
2. (2 points) Calculate Candidate B's predicted lift and drag.

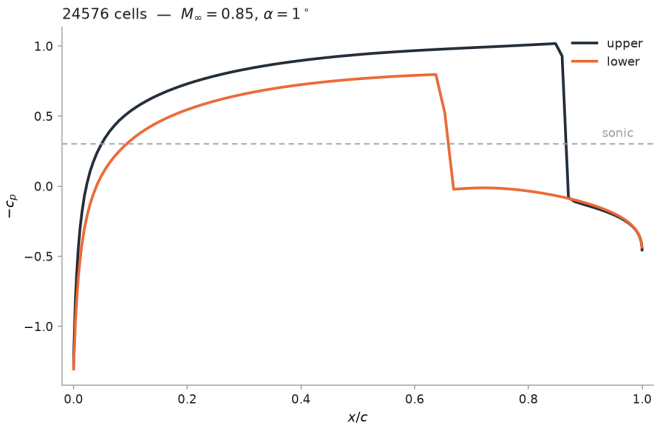
3. (4 points) Complete the compliance matrix.

Candidate	Meets lift requirement?	Meets strong-shock limit?	Keep/reject
A			
B			
C			

4. (2 points) In one sentence each, state why Engineer A's and Engineer B's recommendations are incomplete.

2. Flow-field evidence [7 points]





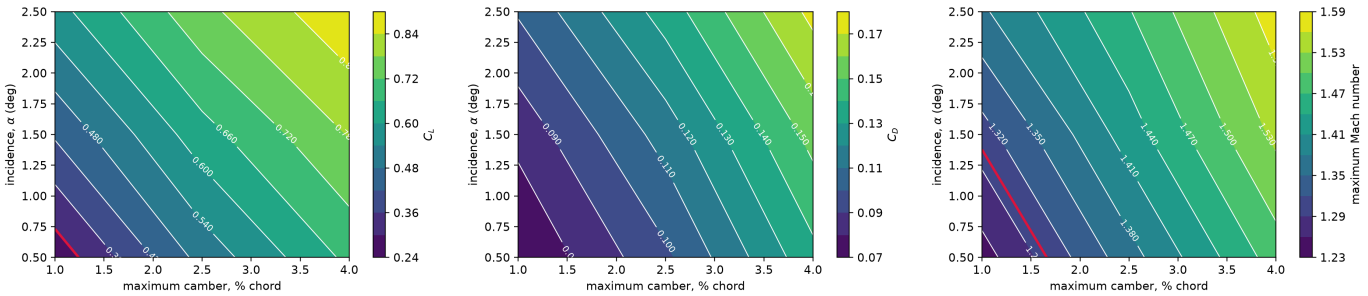
Repository reference case: NACA 0012 at $M_\infty = 0.85$ and $\alpha = 1^\circ$. These figures illustrate the strong-shock mechanism used in the screening constraint.

5. (2 points) On the supplied Mach and C_p plots, mark the shock location.

6. (3 points) Explain how increasing aerodynamic loading can increase lift while also producing a stronger shock and higher drag. Cite one observation from a Mach contour and one observation from a C_p curve.

7. (2 points) Which candidate in the table is most consistent with the strong-shock behavior shown above? State the reason in one sentence.

3. JAX sweep and design recommendation [9 points]



Coarse JAX-adjoint study: fixed 12% thickness and camber crest at $x/c = 0.40$. The red contour marks $C_L = 0.30$ or $M_{\max} = 1.30$.

Candidate B mesh	C_L	C_D
coarse	0.320	0.015
fine	0.322	0.013

8. (2 points) Identify a feasible region of the (m, α) sweep: it must meet both the lift and strong-shock requirements.

9. (2 points) Does the mesh comparison give more confidence in lift or drag? Explain using the table.

10. (5 points) Complete the design record below.

Recommendation: I recommend Candidate _____ / a nearby point in the feasible region because _____.

It meets the payload-weight requirement because _____.

The next simulation or experiment should be _____ because _____.